

1. $5 + x^2 - 3 + 3x$
2. $2y + 5 + 17y + 8$
3. $3x - 6(x - 9)$
4. $6x^3 + 4 + 2x^2(15) + x^2$
5. $4(y + 8) + 3(y - 6)$

$$1. x^2 + 3x + 2$$

$$2. 19y + 13$$

$$3. 3x - 6x + 54 = -3x + 54$$

$$4. 6x^3 + 30x^2 + 4 + x^2 = 6x^3 + 31x^2 + 4$$

$$5. 4y + 32 + 3y - 18 = 7y + 14$$

12. Find the value of $3x^2 + 3(x + 4)$ when $x = 3$.

13. Find the value of $\frac{(x+y)^2}{2} - 10$ when $x = 2$ and $y = 4$.

14. Find the value of $y^2 + 16 - (y - 5)^2$ when $y = 3$.

$$12. 3(3)^2 + 3(3+4) = 27 + 3(7) = 27 + 21 = 48$$

$$13. \frac{(2+4)^2}{2} - 10 = \frac{36}{2} - 10 = 8$$

$$14. (3)^2 + 16 - (3-5)^2 = 9 + 16 - (-2)^2 = 25 - 4 = 21$$

$$11. \frac{5(2x-10)}{2} + 14 = 19$$

$$12. 3(3+r) = 2r + 4$$

$$13. 5y = 2y + 22 + y$$

$$14. 38 = 5(2b-3) + 3b + 1$$

$$15. -5 - x = 2x - (4x + 6)$$

$$16. \frac{3h}{2} = 30$$

$$17. 4(3+2x) + 8 = 92$$

$$18. -5(3-z) = z + 1$$

$$19. 10 - 3b + 3 = -1 + (b + 2)$$

$$20. 5n + 8 - n = 6(n - 1)$$

$$11. \frac{10x-50}{2} = 19-14$$

$$10x - 50 = 2(5)$$

$$10x = 10 + 50$$

$$x = \frac{60}{10}$$

$$x = 6$$

$$12. 9 + 3r = 2r + 4$$

$$3r - 2r = 4 - 9$$

$$r = -5$$

$$13. 5y - 2y - y = 22$$

$$2y = 22$$

$$y = \frac{22}{2}$$

$$y = 11$$

$$14. 38 - 1 = 10b - 15 + 3b$$

$$37 + 15 = 13b$$

$$52 = 13b$$

$$\frac{52}{13} = b$$

$$4 = b$$

$$15. -5 - x = 2x - 4x - 6$$

$$-x - 2x + 4x = -6 + 5$$

$$x = -1$$

$$16. 3h = 2(30)$$

$$h = \frac{2(30)}{3}$$

$$h = 20$$

$$17. 12 + 8x + 8 = 92$$

$$8x = 92 - 8 - 12$$

$$x = \frac{72}{8}$$

$$x = 9$$

$$18. -15 + 5z = z + 1$$

$$5z - z = 1 + 15$$

$$4z = 16$$

$$z = \frac{16}{4}$$

$$z = 4$$

$$19. -3b - b = -1 + 2 - 3 - 10$$

$$-4b = -12$$

$$b = \frac{-12}{-4}$$

$$b = 3$$

$$20. 5n + 8 - n = 6n - 6$$

$$5n - 6n - n = -6 - 8$$

$$-2n = -14$$

$$n = \frac{-14}{-2}$$

$$n = 7$$

Temperature Conversion Formulas

To convert Fahrenheit (F) to Celsius (C)

$$C = \frac{5}{9}(F - 32)$$

To convert Celsius (C) to Fahrenheit (F)

$$F = \frac{9}{5}C + 32$$

23. If the temperature is 68° Fahrenheit, what is the temperature in Celsius?

$$C = \frac{5}{9}(68 - 32) = \frac{5(36)}{9}$$

$$C = \frac{180}{9} = 20^\circ F$$

22. $3x + x = 500$

$4x = 500$

$x = \frac{500}{4}$

$x = 125$

so $E = 3x$

$= 3(125)$

$= 375$

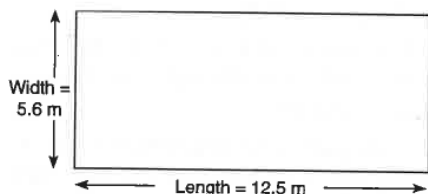
∴ Eric has 375 cards.

Let E be
the number of
cards Eric
has.

22. Dave has 500 baseball cards, which is as many as Eric and Travis have combined. Eric has three times as many cards as Travis has.

Dave	Eric	Travis
500	$3x$	x

From the information, you can write the equation $3x + x = 500$. How many cards does Eric have? (Hint: Solve for x . Then find how many cards Eric has.)



21. Jake has to buy enough fencing to enclose the rectangular garden shown above.

The formula for finding the perimeter of (or distance around) a rectangle is $P = 2l + 2w$, where P = perimeter, l = length, and w = width. Using the values from the drawing, what is the perimeter, in meters, of the garden?

$P = 2L + 2W$

$= 2(12.5) + 2(5.6)$

$= 25 + 11.2$

$= 36.2$

∴ The perimeter of the garden is 36.2 m

$x + 2x + 12 = 174$

$3x = 174 - 12$

$3x = 162$

$x = 54$

∴ Kim earned \$54 on Friday.

24. Kim earned x dollars at his part-time job on Friday. His wife earned \$12 more than twice Kim's pay ($2x + 12$). Together, they earned \$174. How much did Kim earn on Friday? (Hint: Use the equation $x + (2x + 12) = \$174$)

22. If $x = -5$ and $y = 2$, which of the following expressions has the greatest value?

A. $x + y$

B. $-x + y$

C. xy

D. $-2xy$

23. What is the value of the expression $2 \div x^{-4}$ when $x = 2$?

A. 32

B. 16

C. 8

D. -16

22. $x + y = (-5) + 2$
 $= -3$

$-x + y = -(-5) + 2$

$= 7$

$xy = (-5)(2)$

$= -10$

$-2xy = -2(-5)(2)$

$= 20$

23. $\frac{2}{x^{-4}} = 2x^4$

$= 2(2)^4$

$= 2(16)$

$= 32$